US Patent & Trademark Office
Patent Public Search | Text View

United States Patent Application Publication	20250268134
Kind Code	A1
Publication Date	August 28, 2025
Inventor(s)	Walker; Christopher Todd et al.

LATCH ASSEMBLY FOR SECURING BATTERY INSIDE BATTERY HOUSING OF POWER EQUIPMENT

Abstract

A latch assembly for a power equipment includes a lever configured to pivot about a fulcrum, a latch button coupled to the lever and configured to move vertically in response to the pivoting of the lever about the fulcrum, and a roller rotatably engaged to the latch button and adapted to be removably engaged with a battery of the power equipment to removably secure the battery inside a battery housing of the power equipment.

Inventors: **Walker; Christopher Todd (Hillsborough, NC), Desai; Shivang (Portland, OR)**

Applicant: **Honda Motor Co., Ltd.** (Tokyo, JP)

Family ID: **1000007726590**

Appl. No.: **18/584854**

Filed: **February 22, 2024**

Publication Classification

Int. Cl.: **A01D34/81** (20060101); **A01D34/68** (20060101); **A01D34/78** (20060101); **F16B2/20** (20060101); **H01M50/213** (20210101); **H01M50/262** (20210101)

U.S. Cl.:

CPC **A01D34/81** (20130101); **A01D34/68** (20130101); **A01D34/78** (20130101); **F16B2/20** (20130101); **H01M50/213** (20210101); **H01M50/262** (20210101); H01M2220/30 (20130101)

Background/Summary

BACKGROUND

[0001] The disclosed subject matter relates to a latch assembly for a power equipment. More particularly, the disclosed subject matter relates to a latch assembly that facilitates an easy engagement and disengagement of a battery with a battery housing of a power equipment.

[0002] Power equipment, such as lawnmowers, hedge trimmers, string trimmers, power saws, tillers, snow blowers, portable coolers, E-Bikes, and scooters may be operated by an electric motor, or a hybrid of an internal combustion engine and an electric motor. A battery mounted on the power equipment supplies power to the electric motor. The battery is removably mounted onto the power equipment and includes a latch to lock or unlock the battery from a battery housing. However, sometimes, it is difficult to remove the battery from the power equipment, especially in high friction situations, which is undesirable.

SUMMARY

[0003] In accordance with one embodiment of the present disclosure, a latch assembly for a power equipment is disclosed. The power equipment includes a battery housing and a battery removably arranged inside the battery housing. The latch assembly includes a lever configured to pivot about a fulcrum, a latch button coupled to the lever and configured to move vertically in response to the pivoting of the lever about the fulcrum. The latch assembly also includes a roller rotatably engaged to the latch button and adapted to be removably engaged with the battery to removably secure the battery inside the battery housing.

[0004] In accordance with another embodiment of the present disclosure, a power equipment is disclosed. The power equipment includes a battery housing, a battery removably coupled to the battery housing, and a latch assembly configured to removably secure the battery with the battery housing. The latch assembly includes a lever configured to pivot about a fulcrum, a latch button coupled to the lever and configured to move vertically in response to the pivoting of the lever about the fulcrum. The power equipment further includes a roller rotatably engaged to the latch button and adapted to be removably engaged with the battery to removably secure the battery inside the battery housing.

[0005] In accordance with yet a further embodiment of the present disclosure a lawnmower is disclosed. The lawnmower includes a cutter housing including a blade chamber, a plurality of wheels connected to and supporting the cutter housing, a blade arranged at least partially in the blade chamber and rotatably mounted to the cutter housing to rotate about a blade axis, and a battery supported on the cutter housing. The lawnmower further includes a motor mounted on the cutter housing to drive the blade, a lever pivotably supported on the cutter housing, a latch button coupled to the lever and configured to move vertically in response to the pivoting of the lever about the fulcrum, and a roller rotatably engaged to the latch button and adapted to be removably engaged with the battery to removably secure the battery inside the battery housing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] The disclosed subject matter of the present application will now be described in more detail with reference to exemplary embodiments of the apparatus and method, given by way of example, and with reference to the accompanying drawings, in which:

[0007] FIG. **1** is a perspective view of a power equipment, in accordance with one embodiment of the present disclosure;

[0008] FIG. **2** illustrates a perspective schematic view of the power equipment of FIG. **1** with exterior portions of the power equipment shown in phantom and depicting various internal components of the power equipment, in accordance with one embodiment of the present disclosure;

[0009] FIG. **3** illustrates a top perspective view of a latch assembly for selectively securing a battery of the power equipment inside a battery housing of the power equipment, in accordance with one embodiment of the present disclosure;

[0010] FIG. **4** illustrates a top perspective view a latch housing of the latch assembly of FIG. **3**, in accordance with one embodiment of the present disclosure;

[0011] FIG. **5** is a top perspective view of a latch of the latch assembly of FIG. **3**, in accordance with one embodiment of the present disclosure;

[0012] FIG. **6** illustrates a sectional view of the power equipment depicting the latch connected to the battery and retaining the battery inside the battery housing, in accordance with one embodiment of the present disclosure; and

[0013] FIG. **7** illustrates a sectional view of the power equipment depicting the latch disengaged from the battery to enable a removal of the battery from the battery housing, in accordance with one embodiment of the present disclosure.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

[0014] A few inventive aspects of the disclosed embodiments are explained in detail below with reference to the various figures. Exemplary embodiments are described to illustrate the disclosed subject matter, not to limit its scope, which is defined by the claims. Those of ordinary skill in the art will recognize a number of equivalent variations of the various features provided in the description that follows. Embodiments are hereinafter described in detail in connection with the views and examples of FIGS. **1-7**, wherein like numbers indicate the same or corresponding elements throughout the views.

[0015] FIG. **1** illustrates a power equipment **100** according to an embodiment of the disclosure. In the illustrated embodiment, the power equipment **100** is a lawnmower **102**, however, it may be appreciated that the power equipment **100** may be a tiller, a trimmer, snow blowers, portable coolers, e-bike, or any other similar equipment known in the art. The power equipment **100** may extend along an X-direction, a Y-direction and a Z-direction, which directions are orthogonal to each other. As shown, the power equipment **100** includes a plurality of traction members, for example, wheels **104** or tracks, to enable the movement of the power equipment **100** over a surface. Further, the power equipment **100** includes a cutter housing **106** including a blade chamber **108** and supported on the wheels **104**, a handle **110** to enable a holding of the power equipment **100** by a user, and a power source assembly **112**. The cutter housing **106** may be referred to as a mower deck or as a cutter deck or as a deck.

[0016] Referring to FIG. **2**, the power equipment **100** includes a blade **116** rotatably mounted to the cutter housing **106** and arranged at least partially inside the blade chamber **108** and a blade shaft **118** connecting the blade **116** and the power source assembly **112**. The power source assembly **112** is configured to selectively rotate the blade shaft **118** and the blade **116** in the cutter housing **106** about a blade axis **120** which extends in the Z-direction. The power source assembly **112** includes a battery housing **122**, a battery **124** removably arranged inside the battery housing **122**, a blade motor **128** powered by the battery **124** to rotate the blade **116**, a propulsion motor **130** powered by the battery **124**, a transmission **132** operatively connecting the propulsion motor **130** to the rear wheels **104** to operate the wheels **104**. In an embodiment, the transmission **132** may be omitted.

[0017] Referring to FIGS. **2**, **6** and **7**, the battery **124** includes a battery case **136** and a plurality of battery cells **138** arranged within the battery case **136**, and the battery **124** is removably arranged inside a chamber or receptacle of the battery housing **122**. To enable the insertion of the battery **124** inside the battery housing **122** and removal of the battery **124** from the battery housing **122**, the battery housing **122** defines an opening. In some embodiments, a lid may be arranged covering the opening of the battery housing **122**. The opening is arranged/positioned/oriented towards the front end of the power equipment **100** and the battery **124** is inserted into and removed from the battery housing **122** in the Y-direction. Moreover, the battery housing **122** prevents movement of the battery **124** in the X-direction and the Z-direction.

[0018] Additionally, to retain the battery **124** within the battery housing **122** and lock the battery

124 with the battery housing **122** and release the battery **124** from the battery housing **122**, the power equipment **100** includes a latch assembly **140**. As shown in FIGS. **1** and **2**, the latch assembly **140** is disposed proximal to the front end of the cutter housing **106** relative to a rear end of the cutter housing **106** in the Y-direction. As shown in FIGS. **3**, **4**, **6**, and **7**, the latch assembly **140** includes a latch housing **142** having a base **144** and a cover **146** arranged opposite to the base **144** and defining a chamber **148** therebetween, and a latch **150** arranged at least partially inside the latch housing **142** and adapted to be engaged with the battery case **136** to retain the battery **124** within the battery housing **122**. As best shown in FIGS. **3** and **4**, the latch housing **142** is coupled with the battery housing **122**, and the latch housing **142** defines a pair of aligned first grooves **152** and a pair of aligned second grooves **154** extending in the X-direction for supporting the latch **150** that extends in the Y-direction. Although the latch housing **142** is shown to be coupled to the battery housing **122**, it may be envisioned that the latch housing **142** may be integral to the battery housing **122**.

[0019] Referring to FIGS. **3**, **5**, **6**, and **7**, the latch **150** includes a lever **160** that extends in the Y-direction and has a first end **162** arranged proximate to the front end of the battery housing **122** and a second end **164** arranged distally from the front end of the battery housing **122**. Moreover, the latch **150** includes a first pin **166** extending through the lever **160** in the X direction and arranged inside the first grooves **152**. The lever **160** is arranged to rotate/pivot relative to the first pin **166** and rotates about a central axis **168**/rotation axis **168** of the first pin **166**. Accordingly, the first grooves **152** and the first pin **166** together define a fulcrum of the lever **160** about which the lever **160** pivots/rotates between a first position (shown in FIG. **6**) and a second position (shown in FIG. **7**). In the first position, the first end **162** of the lever **160** is arranged vertically downwardly (in the Z-direction) relative to the second end **164** of the lever **160**, while in the second position, the lever **160** is disposed vertically upwardly (in the Z-direction) relative to the second end **164**. It may be appreciated that the lever **160** is biased to the first position, shown in FIG. **5**. For so doing, a biasing member, for example, a spring **182** (best shown in FIGS. **6** and **7**), extends between the latch housing **142** and the latch **150**. For example, the spring **182** extends in a vertical direction with one end connected to the latch **150** at a position proximate to the second end **164** of the lever **160** and other end of the spring **182** is connected to the base **144** of the latch housing **142**. Also, the lever **160** includes a first lever portion **172** extending from the first end **162** to the first pin **166** and a second lever portion **174** extending from the second end **164** to the first pin **166**.

[0020] Moreover, the latch **150** includes a push button **178** arranged at the first end **162** of the lever **160** and engaged with the first lever portion **172** and a latch button **180** disposed at the second end **174** of the lever **160** and connected to the second lever portion **174**. The push button **178** is adapted to be operated by a user to move the lever **160** from the first position to the second position. To enable such a movement of the lever **160**, the push button **178** includes a handle portion **184** adapted to be pushed downwardly in the Z direction by a user, and an engagement portion **186**, for example, a tab **188**, arranged in engagement with the first lever portion **172** of the lever **160** and extending in the Y-direction from the handle portion **184**.

[0021] Moreover, the latch **150** includes a second pin **190** that is arranged at an interface of the handle portion **184** and the engagement portion **186** and extends through the push button **178** in the Y direction and end portions of the second pin **190** are arranged inside the second grooves **154**. Accordingly, the handle portion **184** extends from a first end of the push button **178** to the second pin **190**, while the engagement portion **186** extends from a second end of the push button **178** to the second pin **190**. Also, the engagement portion **186** is arranged vertically below (in Z direction) of the second pin **190** and the handle portion **184**, and supports the first lever portion **172**. Accordingly, the engagement portion **186** is arranged underneath the lever **160** (i.e., first lever portion **172**) contacting the first lever portion **172**.

[0022] The push button **178** is arranged to rotate/pivot relative to the second pin **190** about a central axis/rotation axis **192** of the second pin **190**. Accordingly, the second pin **190** and the second grooves **154** together define a fulcrum about which the push button **178** rotates/pivots. It may be appreciated that the engagement portion **186** moves vertically upwardly in the Z-direction in response to the pushing of the handle portion **184** downwardly in the Z-direction. Due to the movement of the engagement portion **186** in the vertically upward direction, the first lever portion **172** also moves vertically upwardly to the second position, causing the second lever portion **174** of the lever **160** and hence the latch button **180** to move vertically downwardly as the lever **160** rotates/pivots about the fulcrum. As the push force on the handle portion **184** is removed, the handle portion **184** moves vertically upwardly due to the biasing force applied by the spring **182** on the latch button **180**. It may be appreciated that the latch **150** and hence the latch button **180** is disengaged from the battery case **136** as the second end **164** of the lever **160** moves vertically downwardly i.e., in response to the pushing of the handle portion **184** of the push button **178** in the downward direction, while the latch **150** and hence the latch button **180** re-engages with the battery case **136** as the second lever portion **174** and hence the latch button **180** moves upwardly as the downward push force is removed from the handle portion **184**.

[0023] Referring to FIGS. **3**, **5**, **6** and **7**, the latch button **180** includes a body **200** coupled to the second lever portion **174** and extending in the Z-direction. As shown, the body **200** defines a cavity **202** (best shown in FIGS. **5**, **6**, and **7**) and the second end **164** of the lever **160** is arranged/extend inside the cavity **202** to enable coupling of the lever **160** with the body **200**. Accordingly, the latch button **180** is adapted to move in Z direction between a vertically upward position and a vertically downward position in response to the movement of the lever **160** to the first position and the second position, respectively. Therefore, the latch button **180** is biased to the vertically upward position by the spring **182**, and is moved to the vertically downward position by pushing the handle portion **184** of the push button **178** downwardly. It may be appreciated that the latch button **180** is moved to the vertically upward position in response to the rotation/pivoting of the lever **160** in a first direction ‘B’, while the latch button **180** is displaced to the vertically downward position in response to the rotation of the lever **160** in a second direction ‘C’ about the central axis **192**.

[0024] Further, the latch **150** includes roller **208** arranged at a top end of the body **200** and rotatably coupled to the latch button **180**. To facilitate the rotatable coupling of the roller **208** with the latch button **180**, the latch button **180** includes a pair of eye structures **204**, **206** arranged spaced part from each other and facing each other defining a gap therebetween and the roller **208** is arranged inside the gap and is coupled to the eye structures **204**, **206** via a shaft **210** that extends through the roller **208** and the eye structures **204**, **206**. As shown, the shaft **210**, the roller **208**, and the eye structures **204**, **206**, each extend in the X-direction, and the roller **208** is arranged to rotate relative to the shaft **210** about a central axis of the roller **208** or the shaft **210**. Further, to prevent the movement of the shaft **210** in the X-direction relative to the roller **208** and the eye structures **204**, **206**, and to prevent the disengagement of the shaft **210** from the roller **208** and the eye structures **204**, **206**, the latch **150** includes a pair of clips **212** arranged at the ends of the shaft **210**. It may be appreciated that the roller **208** is arranged engaged with the battery case **136** i.e., the battery **124** in the vertically upward position of the latch button **180**, while the roller **208** is arranged outwardly of the battery case **136** i.e., disengaged from the battery case **136** in the vertically downward position of the latch button **180**.

[0025] To facilitate the engagement of the battery **124** i.e., the battery case **136** with the roller **208**, the battery case **136** defines a detent **250** i.e., slot **252**, shown in FIGS. **6** and **7**, that extends in the Z-direction. Therefore, in the vertically upward position of the latch button **180**, shown in FIG. **6**, the roller **208** is arranged inside the slot **252** and is arranged contacting the battery case **136**, retaining and locking the battery **124** within the battery housing **122**. To remove the battery **124** from battery housing **122**, the roller **208** is moved outwardly of the slot **252** by moving the latch button **180** to the vertically downward position, shown in FIG. **7**. For so doing, the user presses the handle portion **184** of the push button **178** downwardly in the Z direction, rotating the push button **178** about the central axis **192** in the first direction ‘B’. Due to the rotation of the push button **178** in the first direction ‘B’ about the rotation axis **192** or the second pin **190**, the engagement portion **186** of the push button **178** moves upwardly in the Z-direction, causing the first lever portion **172** to move upwardly in the Z-direction, thereby rotating the lever **160** about the central axis **168** i.e., rotation axis **168** in the second direction ‘C’. As the lever **160** rotates/pivots in the second direction ‘C’, the second lever portion **174** moves downwardly in the Z direction, causing the latch button **180** to move to the vertically downwardly position, thereby moving the roller **208** out of the detent **250** i.e., slot **252** and disengaging the battery **124** from the battery housing **122** to enable the removal of the battery **124** from the battery housing **122**. As the roller **208** rolls against the surface of the detent **250** i.e., battery case **136**, the friction between the roller **208** and the battery case **136** is reduced, enabling the easy disengagement and removal of the battery **124** from the battery housing **122**.

[0026] Additionally, the push button **178** includes a stopper **254** i.e., a lip **254** arranged connected to the handle portion **184** to restrict a downward movement of the handle portion **184** beyond a certain limit. As the stopper **254** is arranged to contact the cutter housing **106** upon downward movement of the handle portion **184**, the stopper **254** prevents any additional downward movement of the handle portion **184** i.e., the push button **178**.

[0027] It should be noted that the latch assembly as a whole may be oriented in any position. For example, the latch button may move in a horizontal movement (instead of vertical direction) in order to de-latch the battery. Alternatively, the latch button could move diagonally. The mix of terms such as “horizontal” and “vertical” motion should not bind the present disclosure in one specific orientation—each of the buttons and lever may move in various directions (including directions offset from vertical or horizontal directions) and still remain with the scope of the present disclosure.

[0028] In particular, the exemplary X-direction, Y-direction and Z-direction described above may represent a three-dimensional coordinate system for the lawnmower as an exemplary power equipment. However, alternate embodiments of the power equipment may be described by a three-dimensional XYZ coordinate system that is rotated relative to the XYZ coordinate system shown in FIG. **1**. For example, instead of the Z-direction extending in a vertical direction and the X-direction and Y-direction extending in a horizontal direction as shown in FIG. **1**, the XYZ coordinate system may be misaligned with respect to the vertical and horizontal directions.

[0029] Further, alternate embodiments of the latch assembly may be described by a three-dimensional coordinate system that is independent of the three-dimensional coordinate system of the power equipment.

[0030] The lawnmower may be a walk-behind lawnmower, a ride-on lawnmower, a zero-turn-radius lawnmower, a remote-controlled lawnmower, an autonomous lawnmower, or a semi-autonomous lawnmower.

[0031] Instead of inserting and removing the battery from the front of the lawnmower, alternate embodiments of the lawnmower may be configured for insertion/removal of the battery from either side, the rear, or the top of the lawnmower.

[0032] The foregoing description of embodiments and examples has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure to the forms described. Numerous modifications are possible in light of the above teachings. Some of those modifications have been discussed and others will be understood by those skilled in the art. The embodiments were chosen and described in order to best illustrate certain principles and various embodiments as are suited to the particular use contemplated. The scope of the disclosure is, of course, not limited to the examples or embodiments set forth herein, but can be employed in any number of applications and equivalent devices by those of ordinary skill in the art. Rather it is hereby intended the scope of the disclosure be defined by the claims appended hereto.

Claims

1. A latch assembly for a power equipment having a battery housing and a battery removably arranged inside the battery housing, comprising: a lever configured to pivot about a fulcrum; a latch button coupled to the lever and configured to move vertically in response to the pivoting of the lever about the fulcrum; and a roller rotatably engaged to the latch button and adapted to be removably engaged with the battery to removably secure the battery inside the battery housing.
2. The latch assembly of claim 1, wherein the latch button is configured to move between a vertically upward position and a vertically downward position in response to the pivoting of the lever in a first direction and a second direction, respectively, wherein in the vertically upward position, the roller is engaged with the battery, and in the vertically downward position, the roller is disengaged from the battery.
3. The latch assembly of claim 2, wherein the latch button is biased to the vertically upward position.
4. The latch assembly of claim 1 further includes a push button operatively coupled to the lever and adapted to be moved to pivot the lever about the fulcrum to disengage the roller from the battery, wherein the push button and the latch button are arranged on opposite sides of the fulcrum.
5. The latch assembly of claim 4, wherein the latch assembly includes a latch housing and the push button is rotatably coupled to the latch housing and rotates about a rotation axis to pivot the lever about the fulcrum.
6. The latch assembly of claim 5, wherein the push button includes a handle portion adapted to be pushed by a user to rotate the push button in a first direction about the rotational axis, and an engagement portion arranged in engagement with the lever to pivot the lever about the fulcrum in the second direction in response to the rotation of the push button in the first direction.
7. A power equipment, comprising: a battery housing; a battery removably coupled to the battery housing; and a latch assembly configured to removably secure the battery with the battery housing, the latch assembly including a lever configured to pivot about a fulcrum, a latch button coupled to the lever and configured to move vertically in response to the pivoting of the lever about the fulcrum; and a roller rotatably engaged to the latch button and adapted to be removably engaged with the battery to removably secure the battery inside the battery housing.
8. The power equipment of claim 7, wherein the battery defines a detent and the roller is arranged inside the detent to secure the battery with the battery housing and the roller is moved outwardly of the detent to remove the battery from the battery housing.
9. The power equipment of claim 7, wherein the latch button is configured to move between a vertically upward position and a vertically downward position in response to the pivoting of the lever in a first direction and a second direction, respectively, wherein in the vertically upward position, the roller is engaged with the battery, and in the vertically downward position, the roller is disengaged from the battery.
10. The power equipment of claim 9, wherein the latch button is biased to the vertically upward position.
11. The power equipment of claim 7, wherein the latch assembly further includes a push button operatively coupled to the lever and adapted to be moved to pivot the lever about the fulcrum to disengage the roller from the battery, wherein the push button and the latch button are arranged on opposite sides of the fulcrum.
12. The power equipment of claim 11, wherein the latch assembly includes a latch housing and the push button is rotatably coupled to the latch housing and rotates about a rotation axis to pivot the lever about the fulcrum.
13. The power equipment of claim 12, wherein the push button includes a handle portion adapted to be pushed by a user to rotate the push button in a first direction about the rotational axis, and an

engagement portion arranged in engagement with the lever to pivot the lever about the fulcrum in a second direction in response to the rotation of the push button in the first direction.

14. A lawnmower, comprising: a cutter housing including a blade chamber; a plurality of wheels connected to and supporting the cutter housing; a blade arranged, at least partially, in the blade chamber and rotatably mounted to the cutter housing to rotate about a blade axis; a battery housing supported on the cutter housing; a battery removably arranged inside the battery housing; a motor mounted on the cutter housing to drive the blade; a lever pivotably supported on the battery housing; a latch button coupled to the lever and configured to move vertically in response to the pivoting of the lever about the fulcrum; and a roller rotatably engaged to the latch button and adapted to be removably engaged with the battery to removably secure the battery inside the battery housing.

15. The lawnmower of claim 14, wherein the battery defines a detent and the roller is arranged inside the detent to secure the battery with the battery housing and roller is moved outwardly of the detent to remove the battery from the battery housing.

16. The lawnmower of claim 14, wherein the latch button is configured to move between a vertically upward position and a vertically downward position in response to the pivoting of the lever in a first direction and a second direction, respectively, wherein in the vertically upward position, the roller is engaged with the battery, and in the vertically downward position, the roller is disengaged from the battery.

17. The lawnmower of claim 14, wherein the latch button is biased to the vertically upward position.

18. The lawnmower of claim 14 further includes a push button operatively coupled to the lever and adapted to be moved to pivot the lever about the fulcrum to disengage the roller from the battery, wherein the push button and the latch button are arranged on opposite sides of the fulcrum.

19. The lawnmower of claim 18, wherein the push button is rotatably coupled to the battery housing and rotates about a rotation axis to pivot the lever about the fulcrum.

20. The lawnmower of claim 19, wherein the push button includes a handle portion adapted to be pushed by a user to rotate the push button in a first direction about the rotational axis, and an engagement portion arranged in engagement with the lever to pivot the lever about the fulcrum in a second direction in response to the rotation of the push button in the first direction.